

Grammars

Example

Grammar G :

$$S \rightarrow aSb$$
$$S \rightarrow \lambda$$

More Notation

Sentential Form:

A sentence that contains
variables and terminals

Example:

$S \Rightarrow aSb \Rightarrow aaSbb \Rightarrow aaaSbbb \Rightarrow aaabbbb$

Sentential Forms

sentence

We write: $S \stackrel{*}{\Rightarrow} aaabbb$

Instead of:

$$S \Rightarrow aSb \Rightarrow aaSbb \Rightarrow aaasbbb \Rightarrow aaabbbb$$

In general we write: $w_1 \overset{*}{\Rightarrow} w_n$

If: $w_1 \Rightarrow w_2 \Rightarrow w_3 \Rightarrow \cdots \Rightarrow w_n$

By default:

$$w \stackrel{*}{\Rightarrow} w$$

Example

Grammar

$$S \rightarrow aSb$$

$$S \rightarrow \lambda$$

Derivations

*

$$S \Rightarrow \lambda$$

*

$$S \Rightarrow ab$$

*

$$S \Rightarrow aabb$$

*

$$S \Rightarrow aaabbb$$

Example

Grammar

$$S \rightarrow aSb$$

$$S \rightarrow \lambda$$

Derivations

$$S \xRightarrow{*} aaSbb$$

$$aaSbb \xRightarrow{*} aaaaaaSbbbbbb$$

Another Grammar Example

Grammar G : $S \rightarrow Ab$
 $A \rightarrow aAb$
 $A \rightarrow \lambda$

Derivations:

$$S \Rightarrow Ab \Rightarrow b$$

$$S \Rightarrow Ab \Rightarrow aAbb \Rightarrow abb$$

$$S \Rightarrow Ab \Rightarrow aAbb \Rightarrow aaAbbbb \Rightarrow aabbbb$$

More Derivations

$$S \Rightarrow Ab \Rightarrow aAbb \Rightarrow aaAbbb \Rightarrow aaaAbbbb \\ \Rightarrow aaaaaAbbbbbb \Rightarrow aaaaabbbbbbb$$

$$^* \\ S \Rightarrow aaaaabbbbbbb$$

$$^* \\ S \Rightarrow aaaaaaabbbbbbbb$$

$$^* \\ S \Rightarrow a^n b^n b$$

Language of a Grammar

For a grammar G
with start variable S :

$$L(G) = \{w : S \xRightarrow{*} w\}$$

String of terminals

Example

For grammar G : $S \rightarrow Ab$

$$A \rightarrow aAb$$

$$A \rightarrow \lambda$$

$$L(G) = \{a^n b^n b : n \geq 0\}$$

Since: $S \xRightarrow{*} a^n b^n b$

A Convenient Notation

$$\begin{array}{l} A \rightarrow aAb \\ A \rightarrow \lambda \end{array} \quad \longrightarrow \quad A \rightarrow aAb \mid \lambda$$

$$\begin{array}{l} \langle \textit{article} \rangle \rightarrow a \\ \langle \textit{article} \rangle \rightarrow \textit{the} \end{array} \quad \longrightarrow \quad \langle \textit{article} \rangle \rightarrow a \mid \textit{the}$$

Linear Grammars

Linear Grammars

Grammars with
at most one variable at the right side
of a production

Examples: $S \rightarrow aSb$

$$S \rightarrow \lambda$$

$$S \rightarrow Ab$$

$$A \rightarrow aAb$$

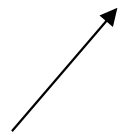
$$A \rightarrow \lambda$$

A Non-Linear Grammar

Grammar G :

$$S \rightarrow SS$$
$$S \rightarrow \lambda$$
$$S \rightarrow aSb$$
$$S \rightarrow bSa$$

$$L(G) = \{w : n_a(w) = n_b(w)\}$$



Number of a in string w

Another Linear Grammar

Grammar G :

$$S \rightarrow A$$
$$A \rightarrow aB \mid \lambda$$
$$B \rightarrow Ab$$

$$L(G) = \{a^n b^n : n \geq 0\}$$

Right-Linear Grammars

All productions have form: $A \rightarrow xB$

or

$$A \rightarrow x$$



Example: $S \rightarrow abS$

$$S \rightarrow a$$

string of
terminals

Left-Linear Grammars

All productions have form: $A \rightarrow Bx$

or

$$A \rightarrow x$$



Example: $S \rightarrow Aab$

$$A \rightarrow Aab \mid B$$

$$B \rightarrow a$$

string of
terminals

Regular Grammars

Regular Grammars

A regular grammar is any
right-linear or left-linear grammar

Examples:

G_1

$$S \rightarrow abS$$

$$S \rightarrow a$$

G_2

$$S \rightarrow Aab$$

$$A \rightarrow Aab \mid B$$

$$B \rightarrow a$$

Observation

Regular grammars generate regular languages

Examples:

G_1

$$S \rightarrow abS$$

$$S \rightarrow a$$

$$L(G_1) = (ab)^* a$$

G_2

$$S \rightarrow Aab$$

$$A \rightarrow Aab \mid B$$

$$B \rightarrow a$$

$$L(G_2) = aab(ab)^*$$

Regular Grammars
Generate
Regular Languages

Theorem

$$\left\{ \begin{array}{l} \text{Languages} \\ \text{Generated by} \\ \text{Regular Grammars} \end{array} \right\} = \left\{ \begin{array}{l} \text{Regular} \\ \text{Languages} \end{array} \right\}$$

Theorem - Part 1

$$\left\{ \begin{array}{l} \text{Languages} \\ \text{Generated by} \\ \text{Regular Grammars} \end{array} \right\} \subseteq \left\{ \begin{array}{l} \text{Regular} \\ \text{Languages} \end{array} \right\}$$

Any regular grammar generates
a regular language

Theorem - Part 2

$$\left\{ \begin{array}{l} \text{Languages} \\ \text{Generated by} \\ \text{Regular Grammars} \end{array} \right\} \supseteq \left\{ \begin{array}{l} \text{Regular} \\ \text{Languages} \end{array} \right\}$$

Any regular language is generated
by a regular grammar

Proof - Part 1

$$\left\{ \begin{array}{l} \text{Languages} \\ \text{Generated by} \\ \text{Regular Grammars} \end{array} \right\} \subseteq \left\{ \begin{array}{l} \text{Regular} \\ \text{Languages} \end{array} \right\}$$

The language $L(G)$ generated by
any regular grammar G is regular

The case of Right-Linear Grammars

Let G be a right-linear grammar

We will prove: $L(G)$ is regular

Proof idea: We will construct NFA M
with $L(M) = L(G)$

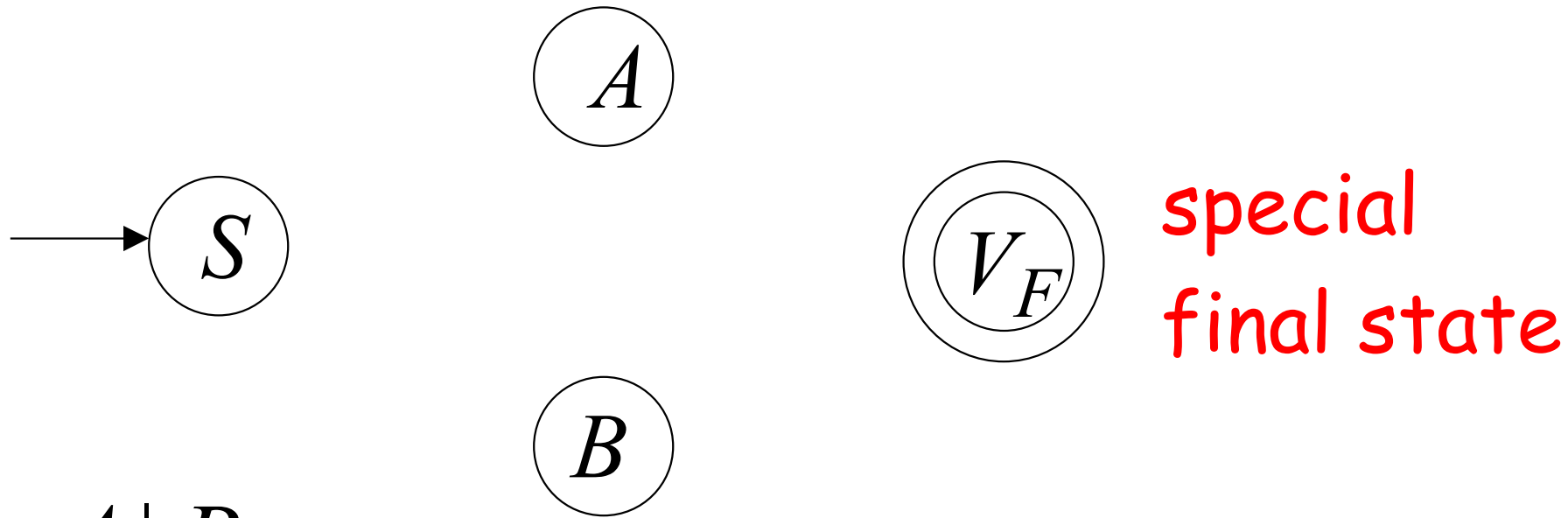
Grammar G is right-linear

Example: $S \rightarrow aA \mid B$

$A \rightarrow aa B$

$B \rightarrow b B \mid a$

Construct NFA M such that every state is a grammar variable:

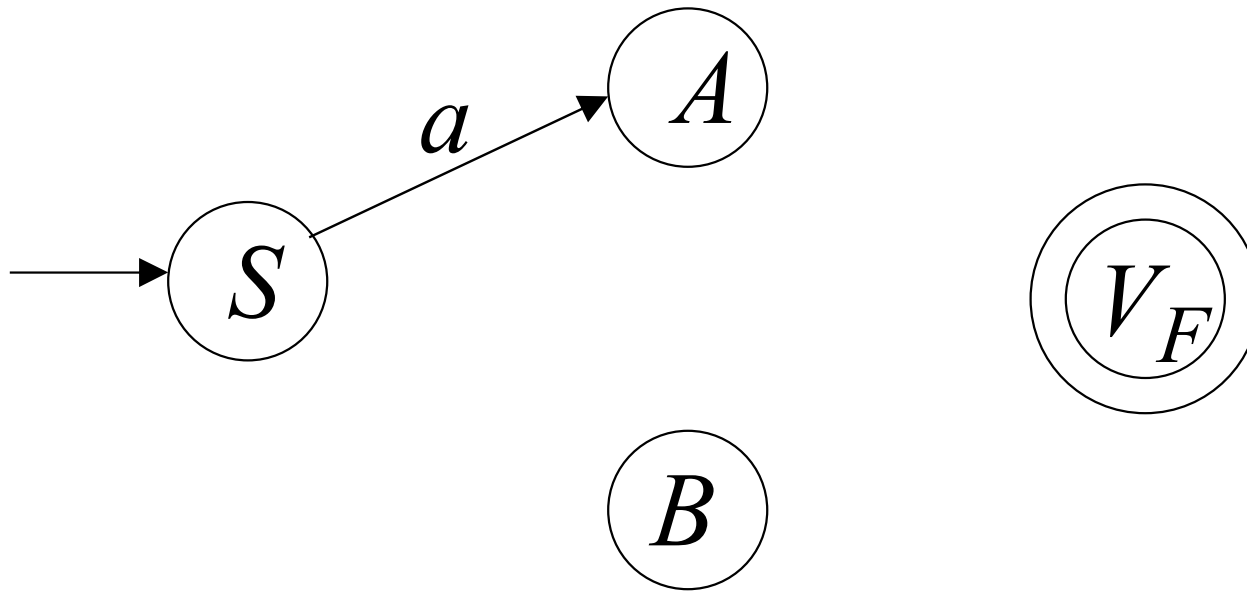


$$S \rightarrow aA \mid B$$

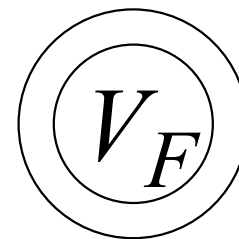
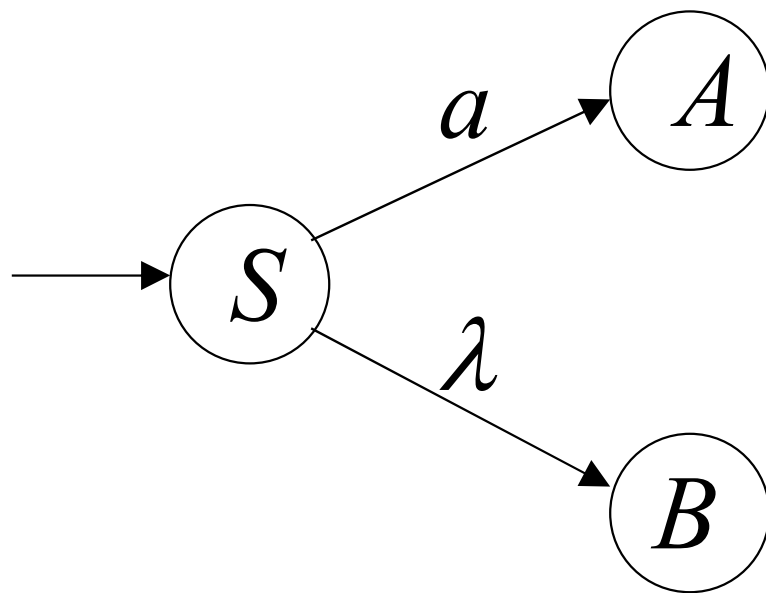
$$A \rightarrow aa B$$

$$B \rightarrow b B \mid a$$

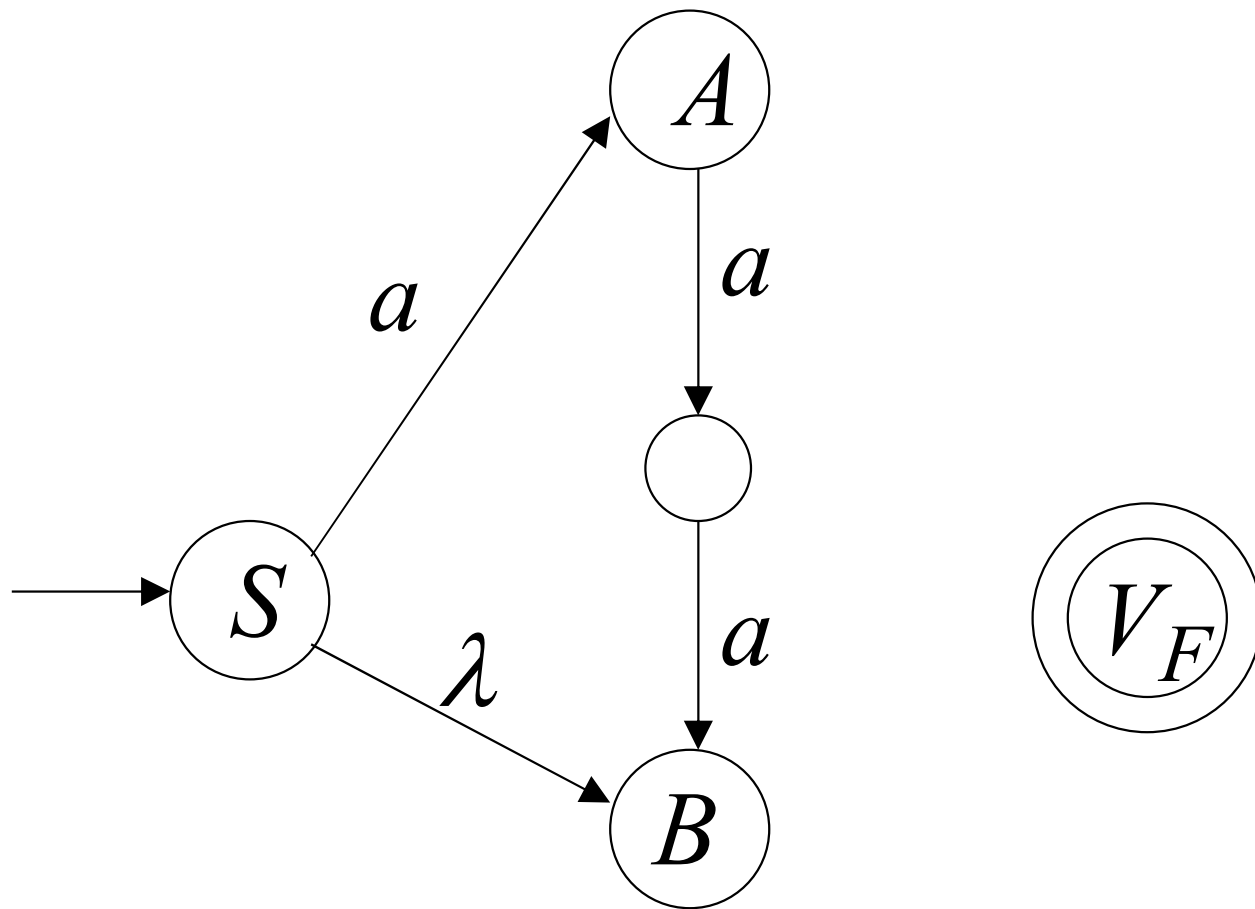
Add edges for each production:



$$S \rightarrow aA$$

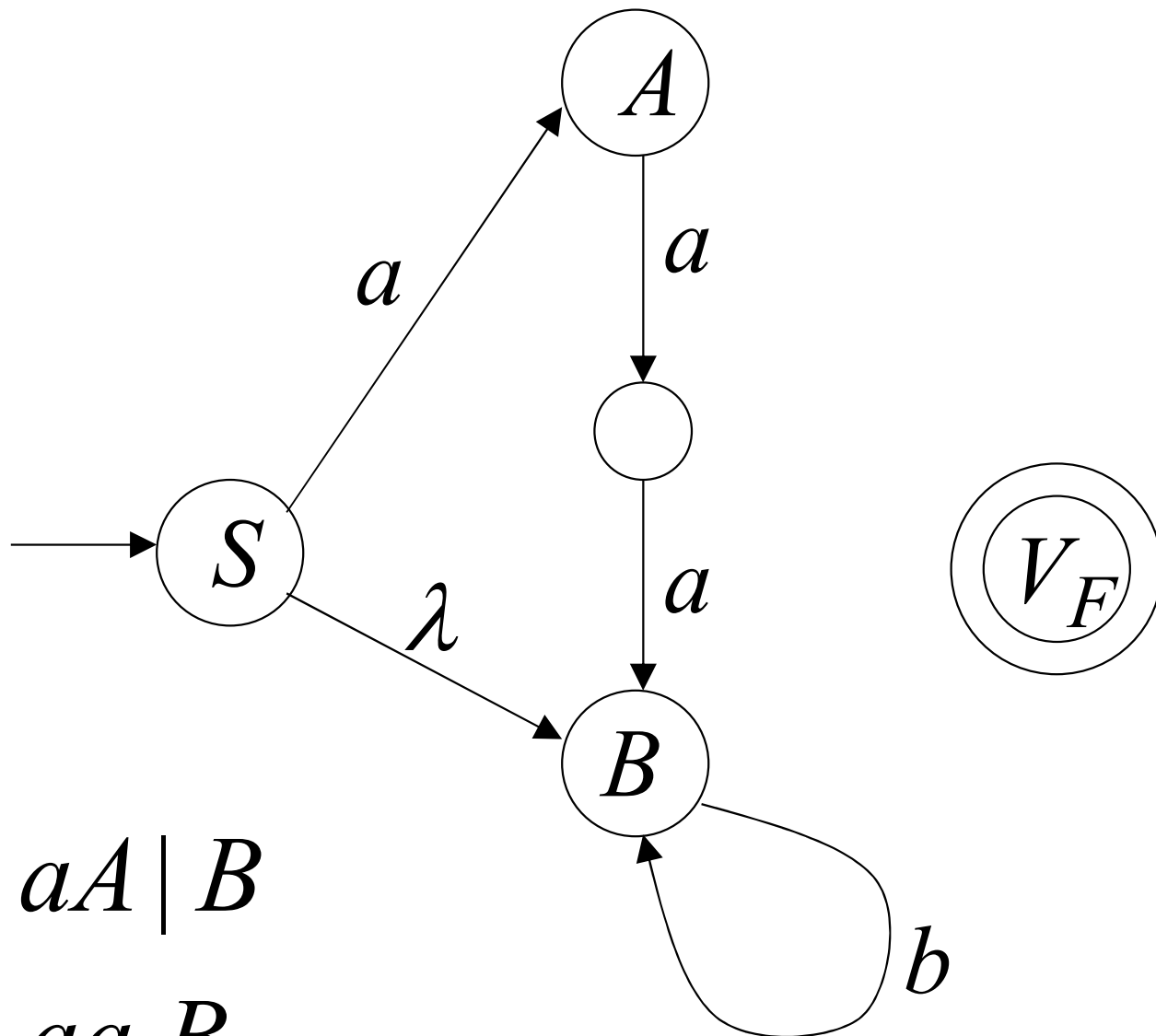


$$S \rightarrow aA \mid B$$



$S \rightarrow aA \mid B$

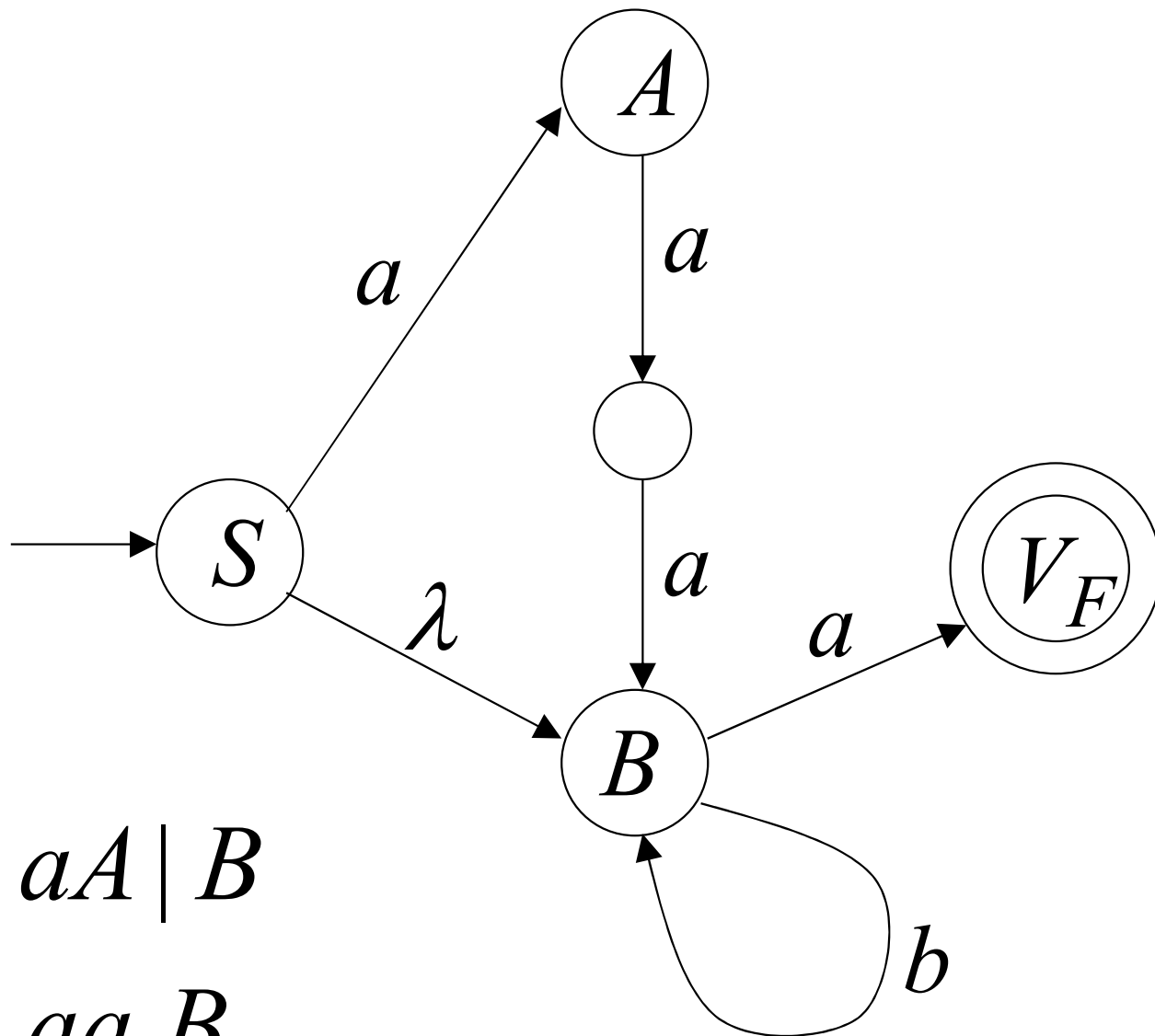
$A \rightarrow aa B$



$$S \rightarrow aA \mid B$$

$$A \rightarrow aa B$$

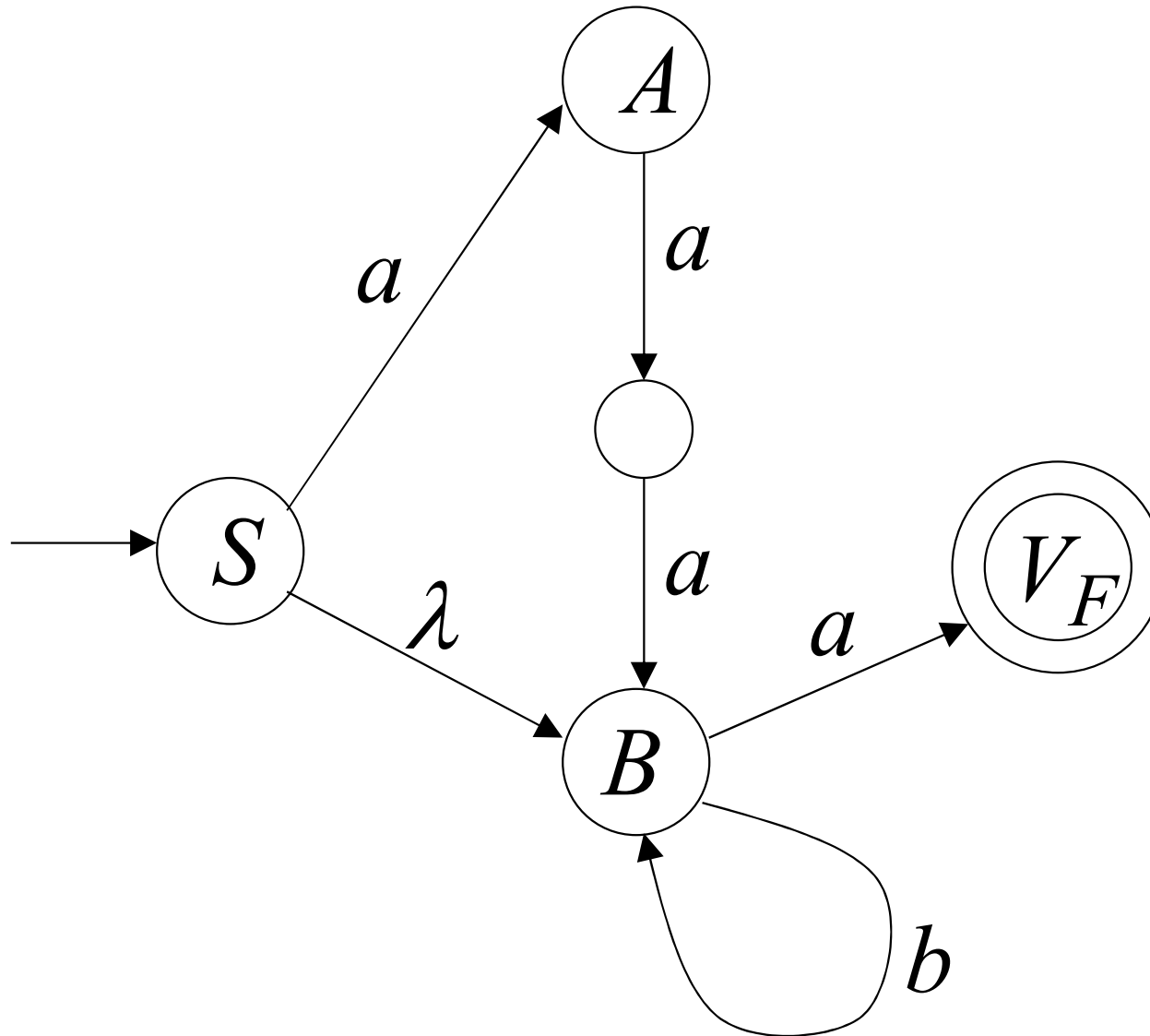
$$B \rightarrow bB$$



$$S \rightarrow aA \mid B$$

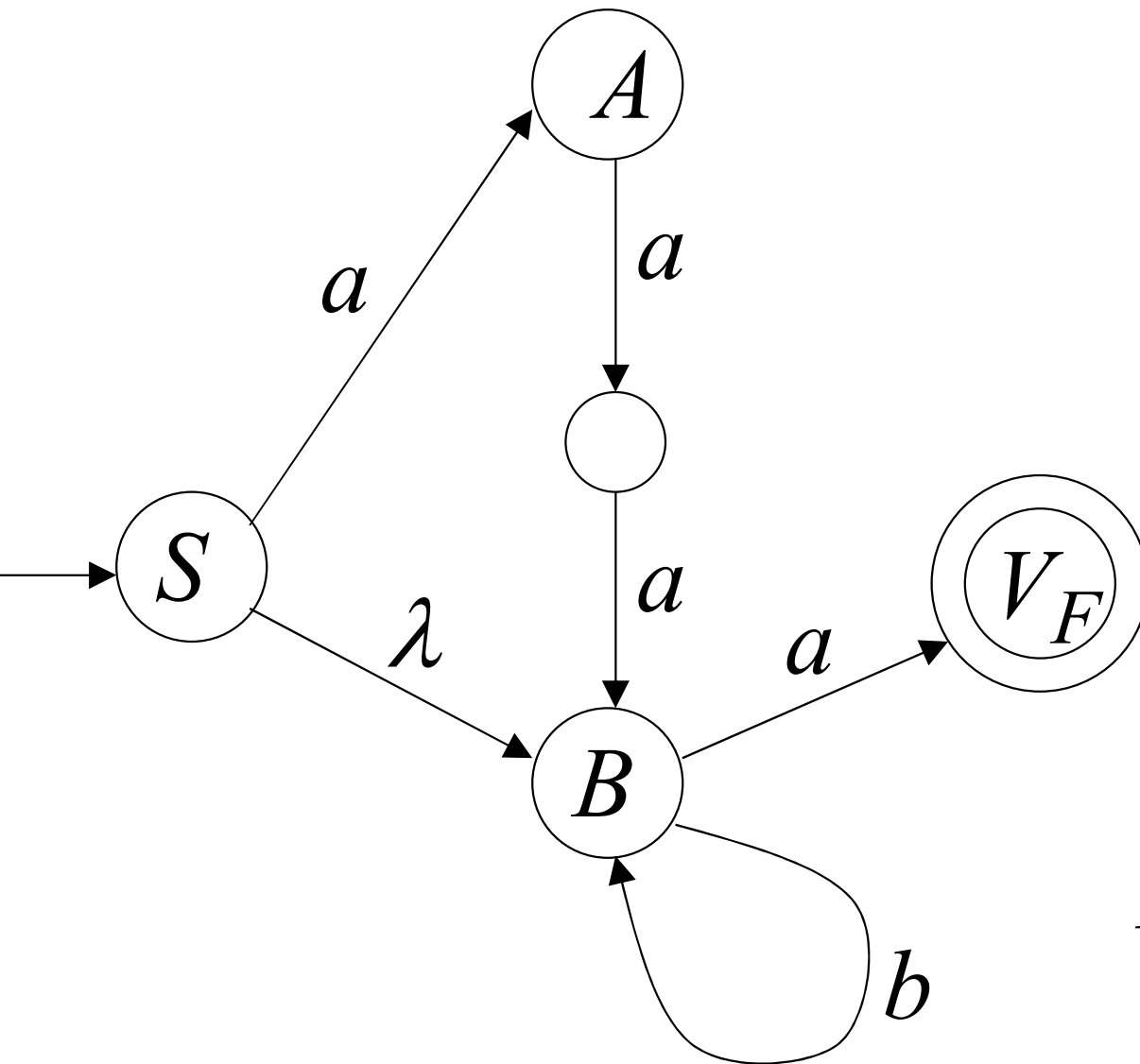
$$A \rightarrow aa B$$

$$B \rightarrow bB \mid a$$



$S \Rightarrow aA \Rightarrow aaaB \Rightarrow aaabB \Rightarrow aaaba$

NFA M



Grammar

G

$S \rightarrow aA \mid B$

$A \rightarrow aaB$

$B \rightarrow bB \mid a$

$$L(M) = L(G) = aaab^*a + b^*a$$

In General

A right-linear grammar G

has variables: $V_0, V_1, V_2, \dots$

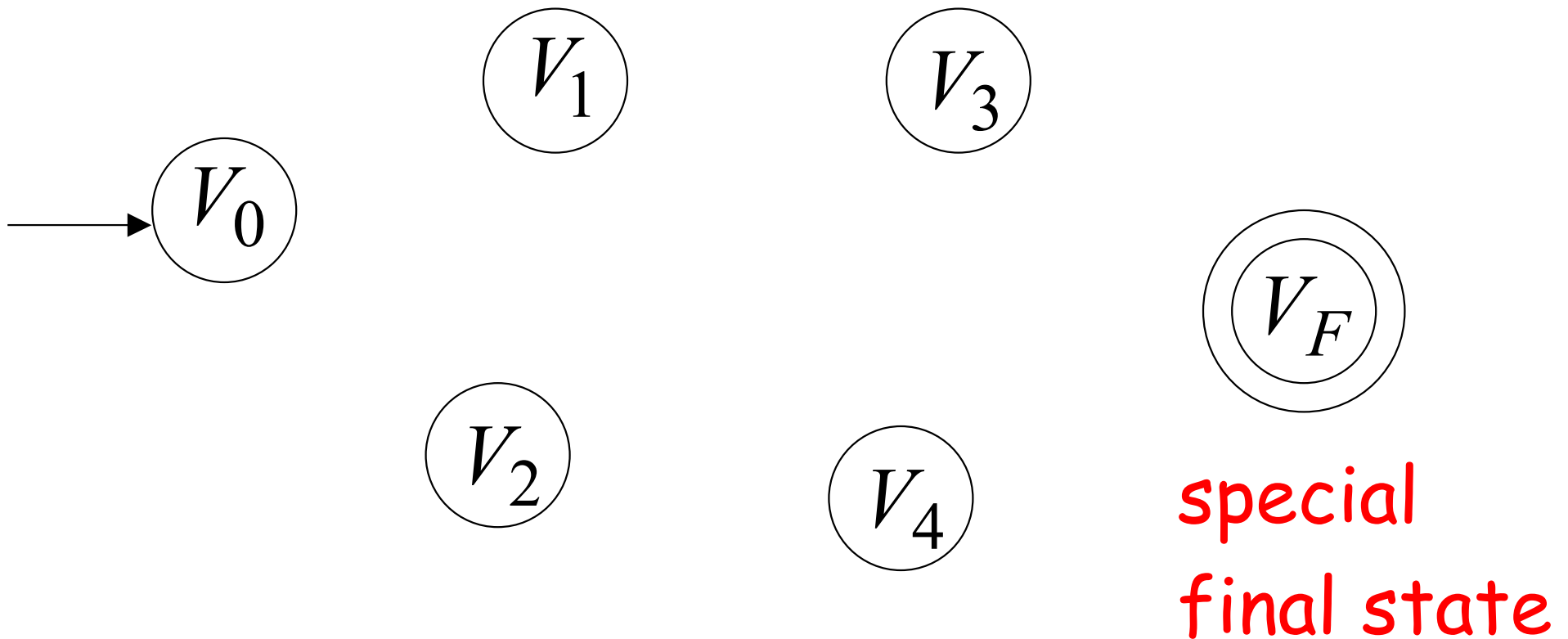
and productions: $V_i \rightarrow a_1 a_2 \cdots a_m V_j$

or

$$V_i \rightarrow a_1 a_2 \cdots a_m$$

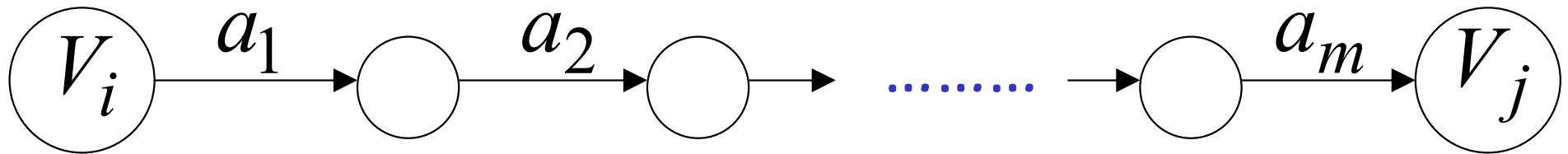
We construct the NFA M such that:

each variable V_i corresponds to a node:



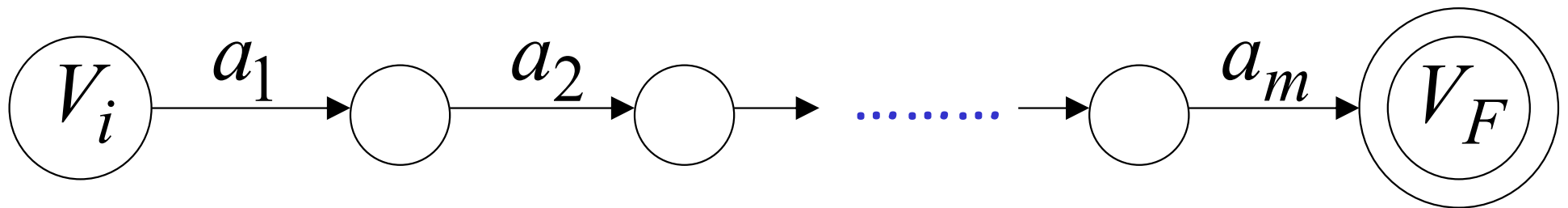
For each production: $V_i \rightarrow a_1 a_2 \cdots a_m V_j$

we add transitions and intermediate nodes

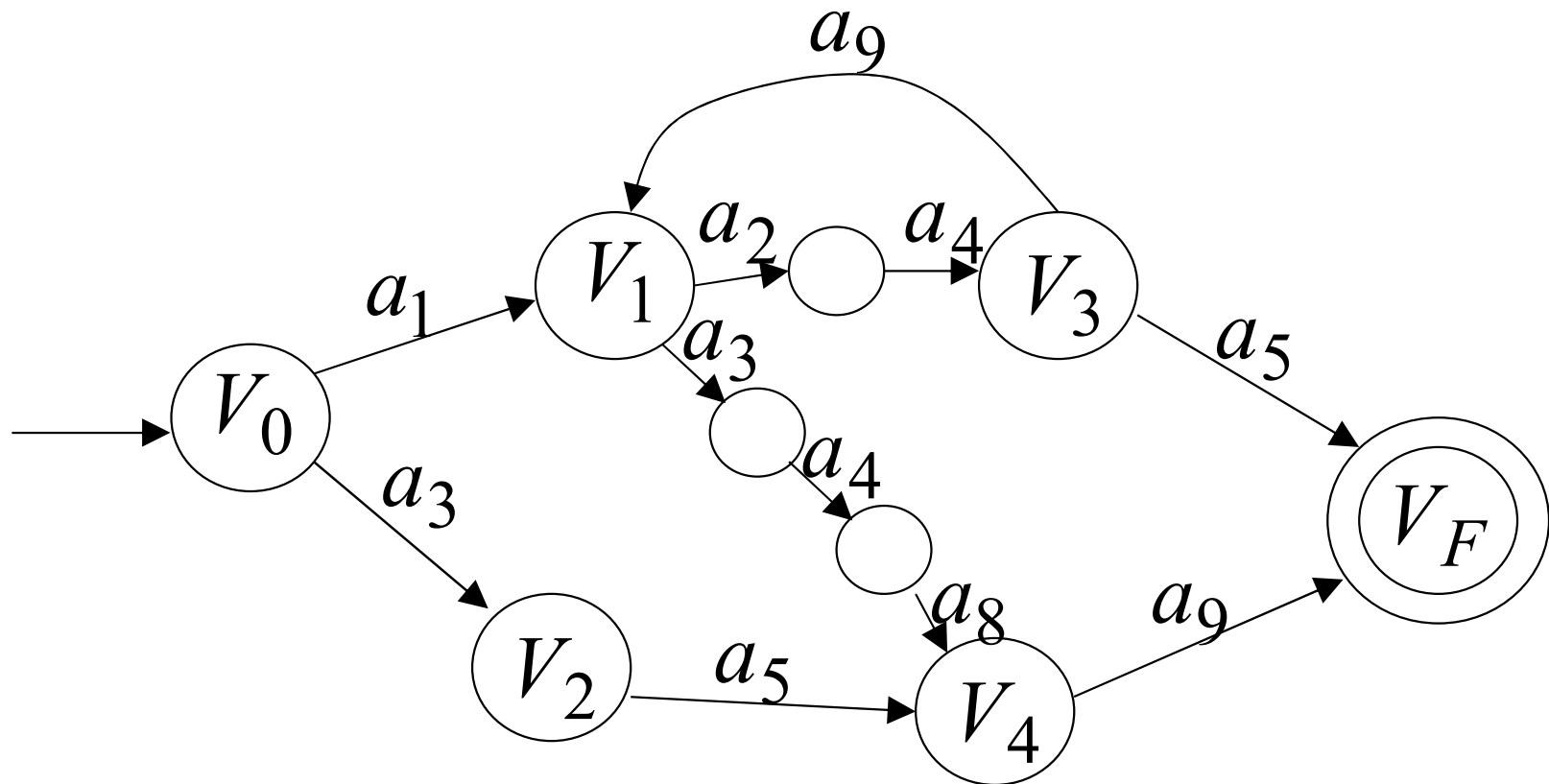


For each production: $V_i \rightarrow a_1 a_2 \cdots a_m$

we add transitions and intermediate nodes



Resulting NFA M looks like this:



It holds that: $L(G) = L(M)$

The case of Left-Linear Grammars

Let G be a left-linear grammar

We will prove: $L(G)$ is regular

Proof idea:

We will construct a right-linear grammar G' with $L(G) = L(G')^R$

Since G is left-linear grammar
the productions look like:

$$A \rightarrow Ba_1a_2 \cdots a_k$$

$$A \rightarrow a_1a_2 \cdots a_k$$

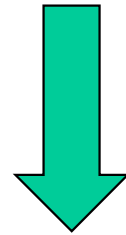
Construct right-linear grammar G'

Left
linear

G

$$A \rightarrow Ba_1a_2 \cdots a_k$$

$$A \rightarrow Bv$$



Right
linear

G'

$$A \rightarrow a_k \cdots a_2a_1B$$

$$A \rightarrow v^R B$$

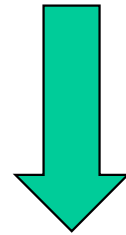
Construct right-linear grammar G'

Left
linear

G

$$A \rightarrow a_1 a_2 \cdots a_k$$

$$A \rightarrow v$$



Right
linear

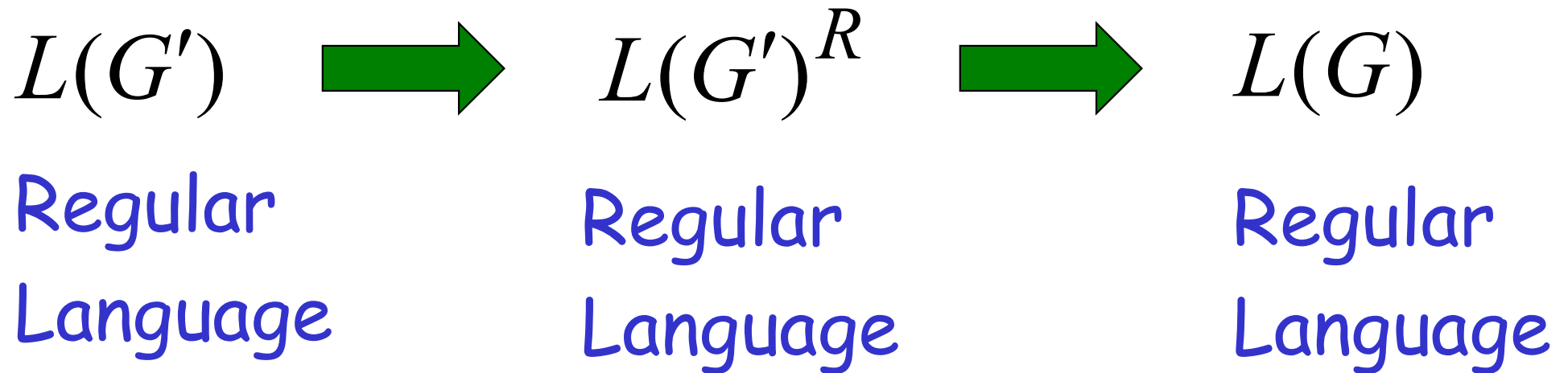
G'

$$A \rightarrow a_k \cdots a_2 a_1$$

$$A \rightarrow v^R$$

It is easy to see that: $L(G) = L(G')^R$

Since G' is right-linear, we have:



Proof - Part 2

$$\left\{ \begin{array}{l} \text{Languages} \\ \text{Generated by} \\ \text{Regular Grammars} \end{array} \right\} \supseteq \left\{ \begin{array}{l} \text{Regular} \\ \text{Languages} \end{array} \right\}$$

Any regular language L is generated
by some regular grammar G

Any regular language L is generated
by some regular grammar G

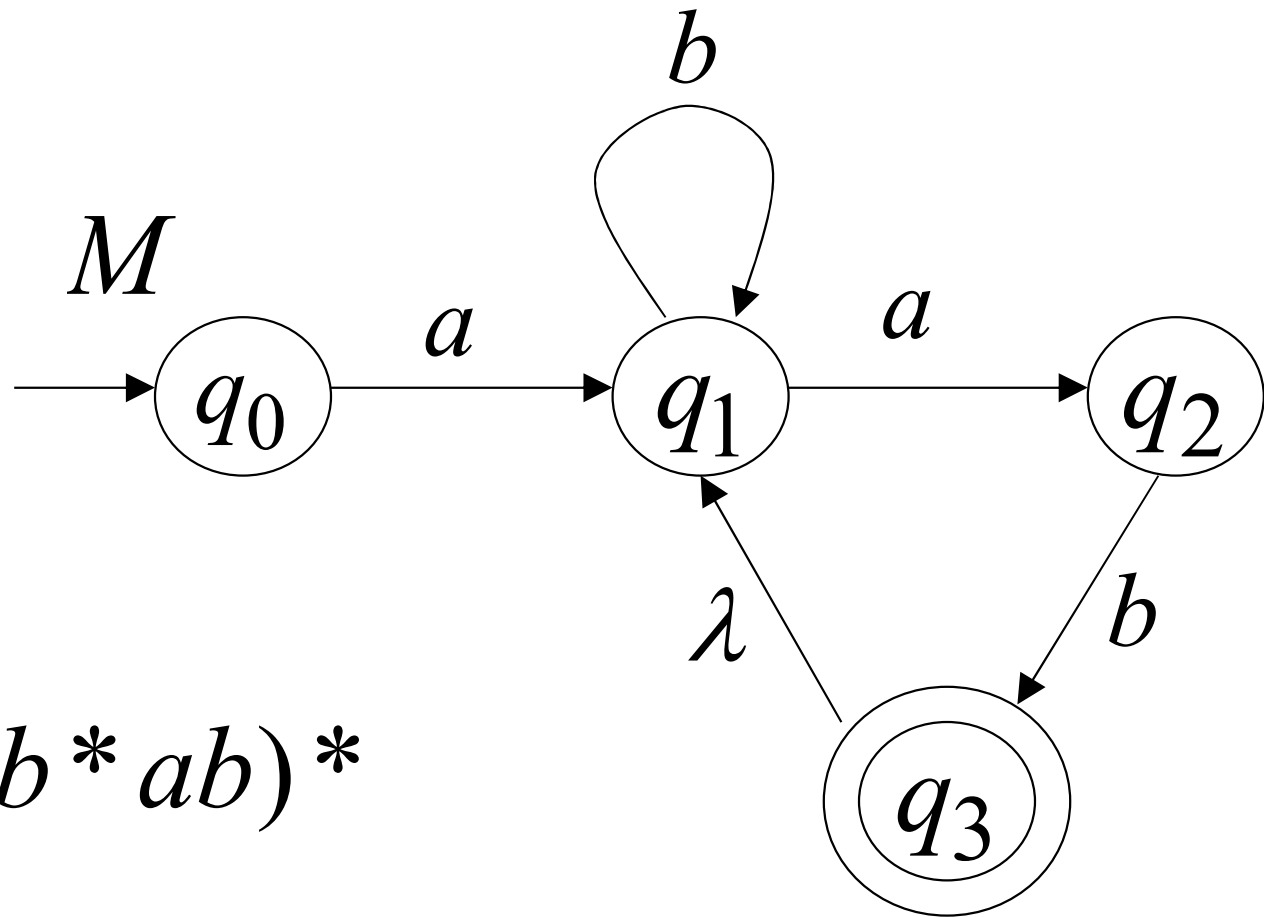
Proof idea:

Let M be the NFA with $L = L(M)$.

Construct from M a regular grammar G
such that $L(M) = L(G)$

Since L is regular
there is an NFA M such that $L = L(M)$

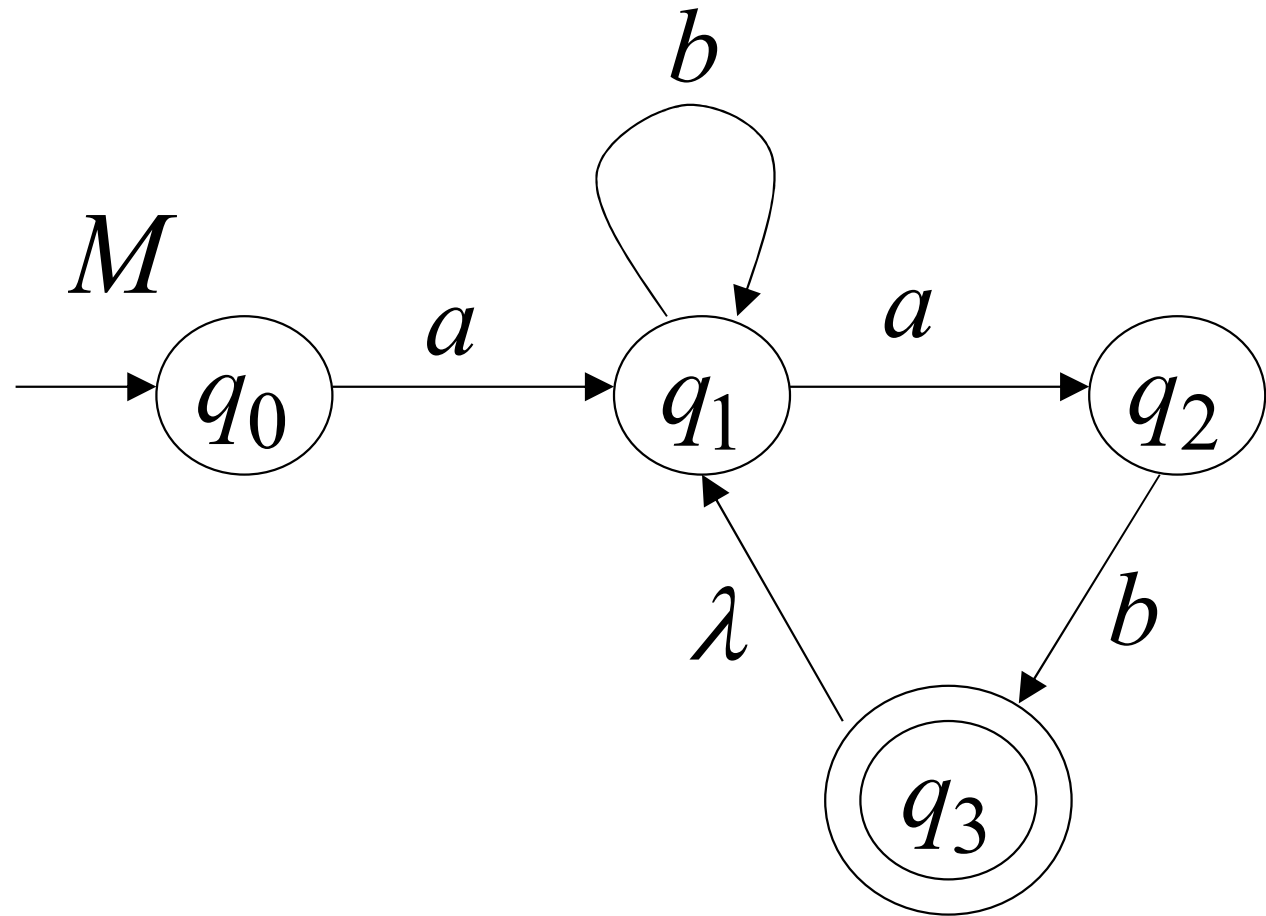
Example:



$$L = ab^*ab(b^*ab)^*$$

$$L = L(M)$$

Convert M to a right-linear grammar

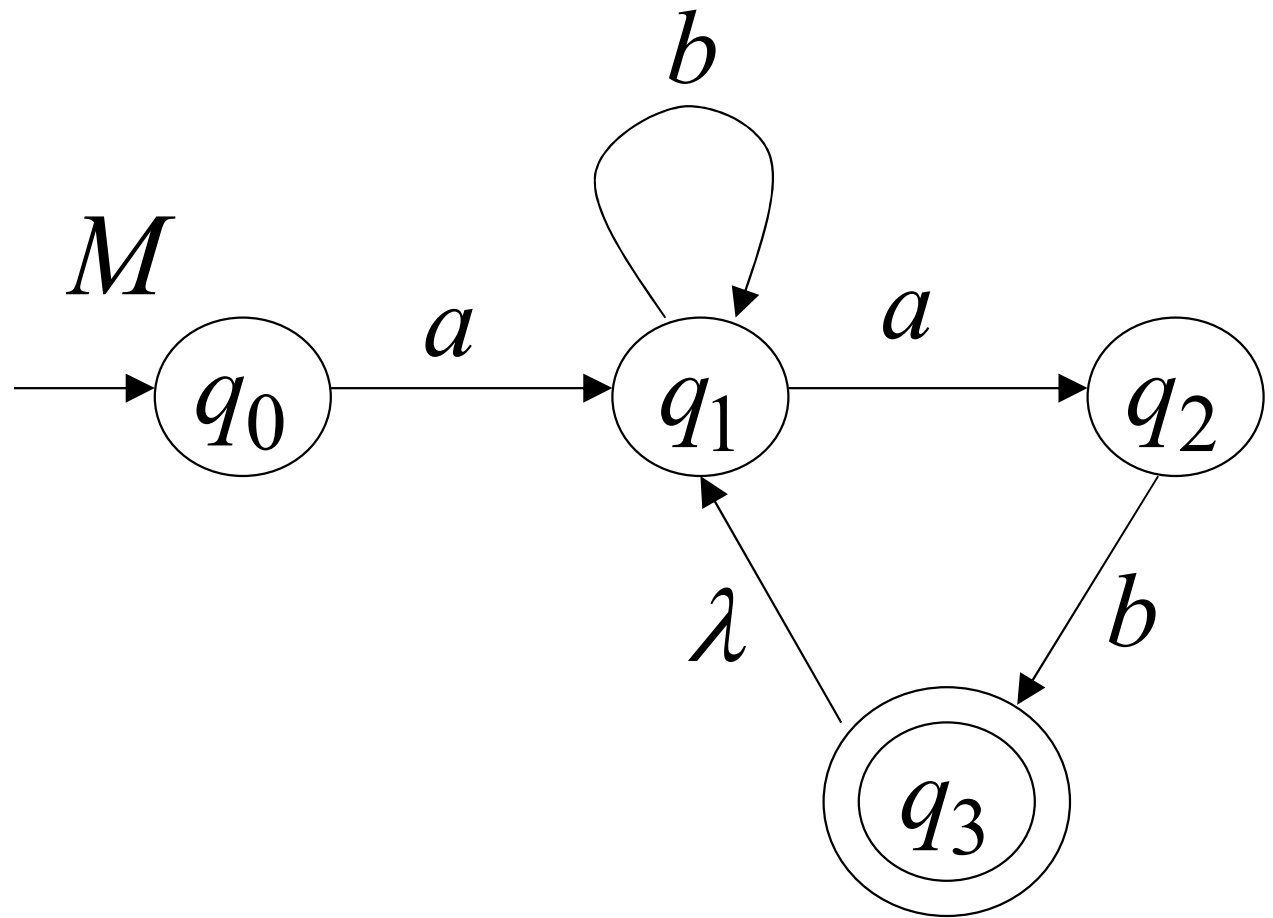


$$q_0 \rightarrow aq_1$$

$$q_0 \rightarrow aq_1$$

$$q_1 \rightarrow bq_1$$

$$q_1 \rightarrow aq_2$$

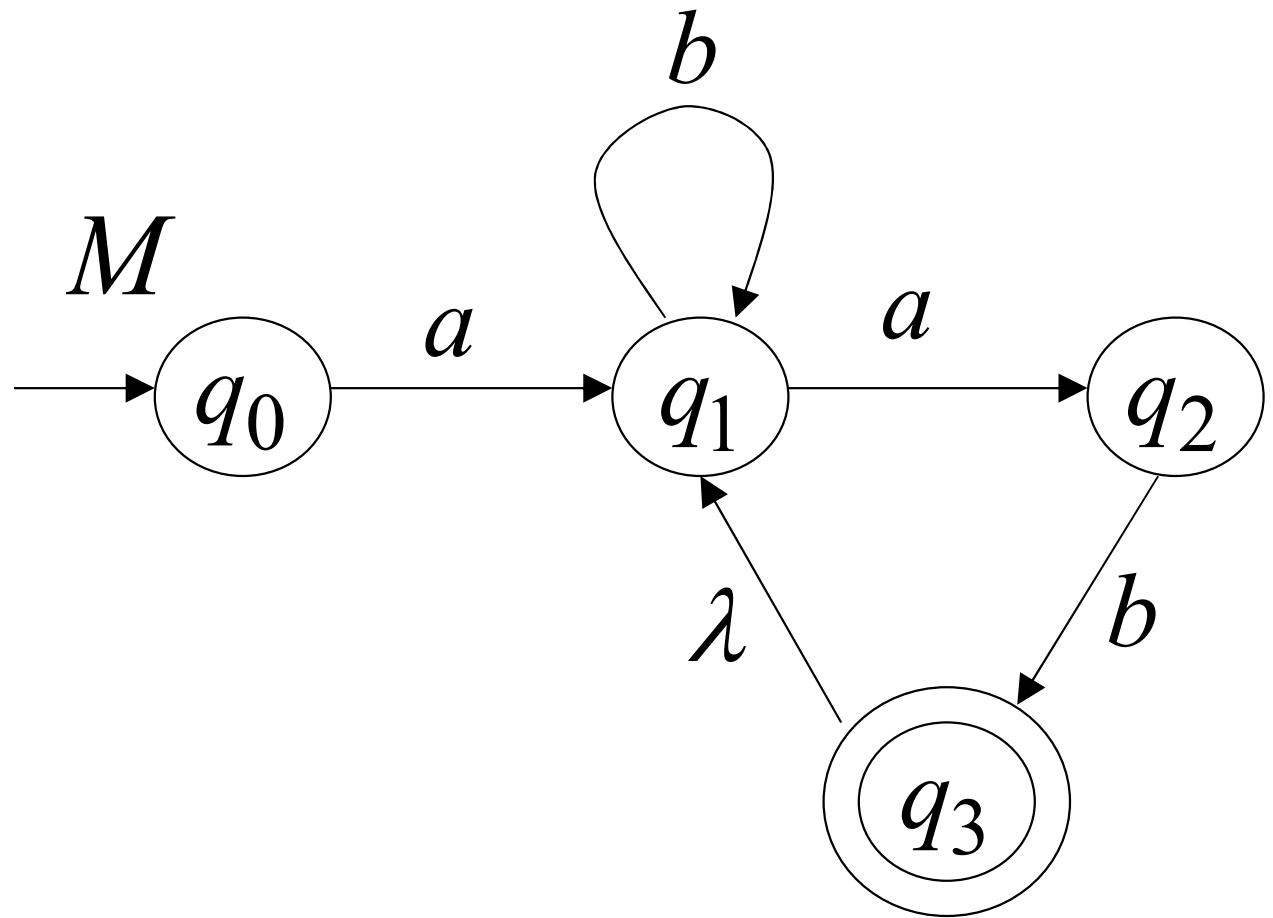


$$q_0 \rightarrow aq_1$$

$$q_1 \rightarrow bq_1$$

$$q_1 \rightarrow aq_2$$

$$q_2 \rightarrow bq_3$$



$$L(G) = L(M) = L$$

G

$$q_0 \rightarrow aq_1$$

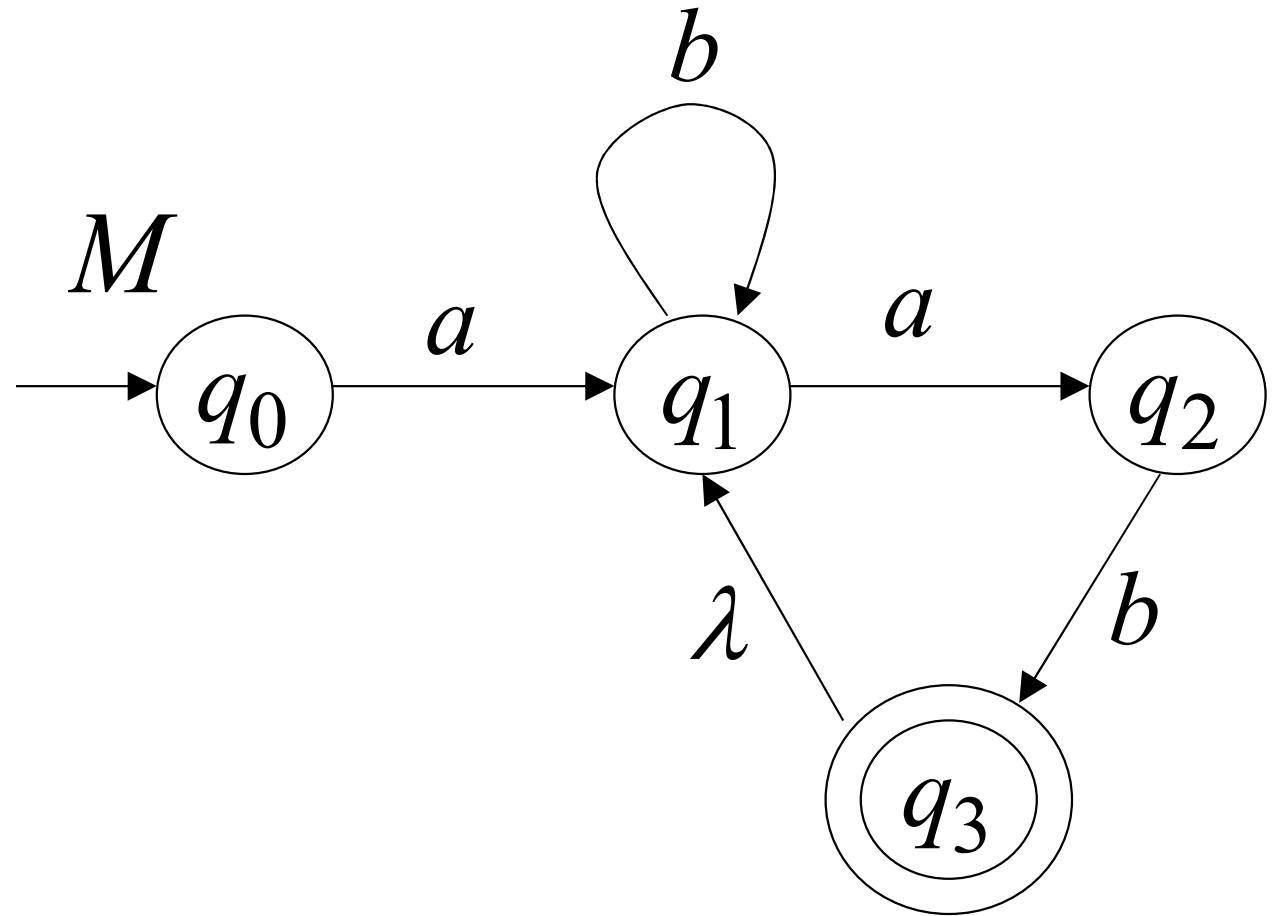
$$q_1 \rightarrow bq_1$$

$$q_1 \rightarrow aq_2$$

$$q_2 \rightarrow bq_3$$

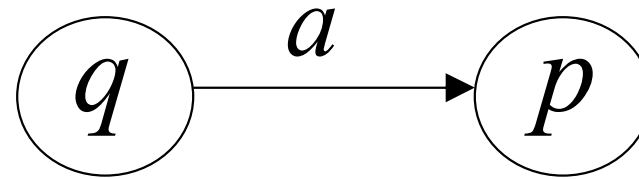
$$q_3 \rightarrow q_1$$

$$q_3 \rightarrow \lambda$$

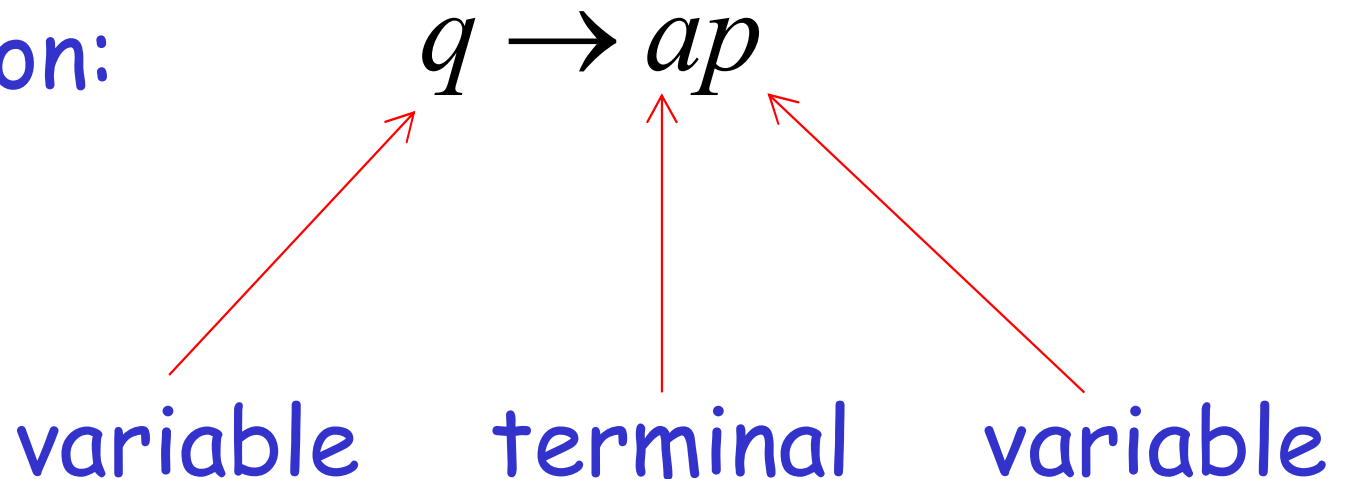


In General

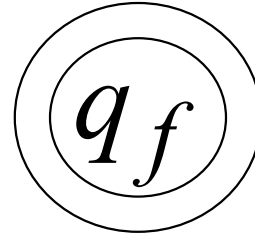
For any transition:



Add production:



For any final state:



Add production:

$$q_f \rightarrow \lambda$$

Since G is right-linear grammar

G is also a regular grammar

with $L(G) = L(M) = L$